Medicare Provider Fraud Detection Using Machine Learning

Technical Report

Demiana Yakoob
Fatma Abdelhadi
Mohammed Abdelsattar
Mohammed Yasser

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Abstract

Healthcare fraud represents a significant financial burden for public insurance programs such as Medicare, costing billions annually. Traditional rule-based fraud detection systems are limited by their inability to generalize to evolving patterns of fraudulent behavior.

This project presents a complete end-to-end machine learning pipeline that integrates multiple Medicare datasets, performs detailed preprocessing and feature engineering, and evaluates several classification models to identify potentially fraudulent providers. The analyses include class imbalance handling, model comparison, feature importance interpretation, and error analysis. The goal is to create an interpretable and scalable fraud-risk scoring system to support real-world investigations.

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1 Introduction

1.1 Motivation

Medicare fraud occurs when healthcare providers submit false or misleading claims for financial gain. Examples include:

- billing for services not provided,
- inflating claim amounts,
- upcoding procedures,
- unnecessary medical services.

Such activities increase national healthcare costs and reduce funds available for genuine patient care.

1.2 Project Goal

We frame fraud detection as a **binary classification problem**:

Given provider-level features, predict whether the provider is fraudulent.

This requires:

1. merging multiple Medicare claims datasets,
2. engineering meaningful provider-level features,
3. choosing and training ML models,
4. evaluating model effectiveness using proper metrics.

2 Data Description

The dataset consists of four primary tables:

- **Provider Table** — provider identifiers + fraud label,
- **Inpatient Claims**,
- **Outpatient Claims**,
- **Beneficiary Table**.

Each table contributes essential information that must be aggregated to the provider level.

2.1 Target Variable

Fraud label:

- `1` = fraudulent provider,
- `0` = non-fraudulent provider.

Figure 1: TODO: Insert class distribution plot showing fraud vs non-fraud.

3 Data Preprocessing Pipeline

This section describes the complete steps applied before training models.

3.1 1. Merging and Relational Mapping

We join tables using:

- **ProviderID** to link claims → providers,
- **BeneID** to retrieve patient demographics.

All claims belonging to the same provider are aggregated.

3.2 2. Cleaning and Standardization

- Removal of invalid identifiers.
- Formatting date fields.
- Handling missing demographic values (imputation with median/mode).
- Dropping duplicate records.

3.3 3. Outlier Detection

To reduce skewed billing patterns, we used:

- IQR-based outlier capping for numerical features,
- log-transform for extremely right-skewed cost variables.

3.4 4. Encoding Categorical Features

Examples:

- Diagnosis codes → count encodings,
- Procedure codes → frequency encodings.

3.5 5. Final Feature Consolidation

We aggregate the following provider-level metrics:

- claim count features,
- average procedure cost,
- number of unique beneficiaries,
- patient chronic condition ratio,
- total vs average claim amounts,
- distribution of inpatient vs outpatient claims.

Figure 2: TODO: Insert complete preprocessing pipeline diagram.

4 Feature Engineering and Analysis

4.1 Summary Statistics

We explore:

- distribution of claim amounts,
- number of beneficiaries per provider,
- code frequency patterns,
- chronic condition profiles.

Figure 3: TODO: Insert correlation heatmap or EDA visualizations.

4.2 Feature Importance Expectation

We hypothesize that fraudulent providers will have:

- unusually high claim frequency,
- abnormal billing patterns (mean and variance),
- unusual diagnosis and procedure code combinations.

5 Class Imbalance Strategy

Fraud cases typically represent less than 10% of providers $\rightarrow$ models tend to predict only the majority class.

To address this, we used:

- **Class-weighted models** (Logistic Regression, Random Forest),
- **SMOTE oversampling** (optional),
- Evaluation using precision, recall, F1, and AUC instead of accuracy.

6 Modeling Approach

6.1 Models Evaluated

1. Logistic Regression
2. Random Forest Classifier
3. XGBoost Classifier

Each model was tuned using cross-validation and hyperparameter search.

6.2 Train/Test Split

We used a stratified split:

- 80% training,
- 20% testing.

7 Results and Analysis

7.1 Model Comparison Table

Table 1: Model performance summary (replace with actual results).

Model	Precision	Recall	F1-score	AUC
Logistic Regression	0.XX	0.XX	0.XX	0.XX
Random Forest	0.XX	0.XX	0.XX	0.XX
XGBoost	0.XX	0.XX	0.XX	0.XX

7.2 ROC Curve

Figure 4: TODO: Insert ROC curve for all models.

7.3 Confusion Matrix

Figure 5: TODO: Insert confusion matrix for best model.

7.4 Feature Importance

Figure 6: TODO: Insert feature importance plot.

7.5 Interpretation

Example interpretation (replace with your notebook insights):

- Providers with **high average claim cost** show higher fraud likelihood.
- **Frequent procedure repetition** signals abnormal patterns.
- **Large beneficiary counts** with high-cost claims indicate potential fraud schemes.

8 Discussion

8.1 Key Findings

- Ensemble models outperform linear models.
- Fraud is strongly associated with abnormal utilization patterns.
- Explainability tools (feature importance, partial dependence) help interpret predictions.

8.2 Operational Use-Case

Model predictions can be used to:

- rank providers by fraud likelihood,
- prioritize audit resources,
- reduce financial losses through early detection.

9 Limitations and Future Work

9.1 Limitations

- Claims lack temporal sequence modeling.
- Fraud labels may contain noise or incomplete investigations.
- Features may not capture entire behavioral context.

9.2 Future Work

- Incorporate time-series models (LSTM, Transformers).
- Apply SHAP for explainable AI.
- Build an interactive fraud scoring dashboard.

10 Conclusion

This project demonstrates a complete machine learning approach for Medicare fraud detection, including data integration, preprocessing, feature engineering, modeling, and evaluation. Ensemble models show the highest predictive performance and provide actionable insights for fraud investigation units. With further enhancements, this system can support scalable and cost-effective fraud detection nationwide.